

V2026.1 ENTERPRISE ARCHITECTURE

MarketMind AI

Autonomous Multi-Agent Market Research Loop

An enterprise-grade research automation engine leveraging specialized LLM micro-agents, deterministic tool dispatching, epistemic classification, and strict cost controls.

Executive Summary & Core Objectives

Transforming manual desk research into verifiable, automated intelligence

01. Autonomous Execution

Eliminates manual desk research through recursive self-directed planning, multi-tool querying, and structured evidence accumulation.

02. Micro-Specialization

Decomposes research pipelines into dedicated role-based agents (Planner, Researcher, Synthesis, Quality Control, Evidence Analyst).

03. Quality & Human Gate

Enforces strict epistemic classification (Fact vs Inference) backed by automated quality controls and interactive human sign-off gates.

Primary Value Proposition

Reduces market research turnaround time from days to minutes while maintaining auditability and strict source attribution for every single claim.

Deterministic Reliability

Guaranteed token budget ceilings, max loop iteration caps, and automated hallucination filtering via strict tool result validation.

Multi-Agent System Architecture

End-to-end orchestration flow across specialized LLM agents and execution components

STEP 1

Planner Agent

Parses user research request into a structured execution plan with clear scope and targeted objectives.

STEP 2

Researcher Loop

Generates search queries, executes tool calls, gathers raw web data, and populates evidence store.

STEP 3

Synthesis & QC

Aggregates evidence, performs epistemic claim classification, and verifies findings against limitations.

STEP 4

Human Approval

Presents interactive dashboard with sign-off options: Approve, Rescope, Expand, or Reject.

Core Dispatcher Engine: Tool execution is handled via deterministic Pydantic schemas, routing requests to live API handlers with fallback handling and token tracking.

Agent Micro-Specialization

Dedicated roles prevent prompt bloat and improve decision precision

Agent Module	Core Role & Responsibility	Input Data	Output Artifact
Planner Agent	Decomposes prompt into logical research sub-goals	User Prompt / Constraints	Structured Research Plan JSON
Researcher Agent	Executes queries, parses raw web data & API calls	Research Plan & State	Raw Web Tool Results
Evidence Analyst	Extracts claims, validates citations & source text	Raw Tool Data	Normalized Evidence Records
Synthesis Agent	Drafts comprehensive brief & classifies claims	Evidence Records	Draft Intelligence Report
Quality Control	Identifies gaps, flags uncertainties, checks bounds	Draft Report & Rules	QC Status & Audit Log

Epistemic Claim Classification & Evidence Store

Strict categorization ensures full transparency and eliminates unverified statements

FACT

Directly verifiable data backed by explicit, primary source citations (e.g., subscription pricing tier tables).

INFERENCE

Logical deductions derived from combining multiple verified factual claims (e.g., market shift trends).

REC

Strategic action recommendations formulated based on analytical findings and user objectives.

UNCERTAIN

Explicitly flagged knowledge gaps or missing data points (e.g., unlisted custom enterprise pricing).

Traceability Matrix: Every claim emitted in the final report contains a unique `source_ref` pointing directly to an immutable record in the Evidence Store containing source URL, retrieval timestamp, and raw snippet context.

Human-in-the-Loop Approval Gate

Interactive dashboard control preventing autonomous drift

APPROVE BRIEF

Signs off on the current research report and triggers export to PDF/Markdown.

REJECT

Terminates execution loop and discards output due to alignment failures.

EXPAND SCOPE

Instructs agents to perform deeper analysis on identified uncertainties or gaps.

MODIFY SCOPE

Allows human operator to adjust parameters, pricing constraints, or targets.

Built-in Web Interface

Available both as a Streamlit dashboard (`app.py`) and a standalone React single-page frontend (`index.html`) with Tailwind styling.

Performance & Cost Efficiency Benchmarks

Predictable, cost-capped execution vs traditional desk research

Metric / Parameter	Manual Desk Research	MarketMind AI Autonomous Loop	Improvement Factor
Turnaround Time	8 to 24 Hours	45 to 120 Seconds	~100X FASTER
Cost per Brief	\$200.00 - \$500.00 (Human)	\$0.25 - \$0.85 (LLM API)	~500X LOWER
Source Attribution	Manual bookmarks / Notes	100% Traceable Evidence Store	DETERMINISTIC
Epistemic Rigor	Subjective reporting	Strict Fact/Inference Split	AUDIT READY

Deployment & Quickstart Setup

Getting started with MarketMind AI in 3 simple steps

1. Setup Environment

```
git clone marketmind.git
cd marketmind
python -m venv venv
pip install -r requirements.txt
```

2. Configure Keys

```
cp .env.example .env
# Set your OpenAI key:
OPENAI_API_KEY=sk-proj-...
```

3. Launch Interface

```
# Launch Streamlit UI
streamlit run app.py

# Or run smoke test
python smoke_test.py
```

Ready for Production: Complete with automated pytest suites, evaluation runners, and structured JSON logging.